Wafer Defect Detection and Classification using WM811k

Dataset Analysis:

Imbalance Analysis: The distribution in Figure 1 shows a severe class imbalance in the dataset. The majority class (None, representing wafers without specific defect patterns) contains significantly more samples than the combined number of actual defect classes. For instance, classes like Scratch and Near-full have extremely low representation. Training a Convolutional Neural Network (CNN) directly on this raw distribution would result in a biased model that over-predicts the None class to achieve superficially high accuracy. To counter this and build a robust classifier, our pipeline requires specialized techniques. We employ class weights during loss calculation (penalizing errors on rare defects more heavily) and use data augmentation/resampling techniques to ensure the model learns the geometric features of all 9 failure types equally.

Extreme class: For the WM811k wafer map classification problem, analysis must address the extreme class imbalance and the spatial nature of defect patterns to prove your model is reliable. The dataset consists of over 811,000 wafer maps, but only a small fraction are labeled, with nine main classes (8 defect types + None).

End-to-end data Preprocessing: We need to address several critical areas before performing the classification analysis. They are described as follows.

A) The script must ensure the data is clean and uniform. B) Defect datasets are notoriously imbalanced. To minimize this effect, down-sampling in the majority class (here, the dominant non-class, i.e., wafers with no defects) is used. C) Advanced Augmentation for underrepresented defect types, the script creates synthetic examples by applying rotations and flips. Further, the

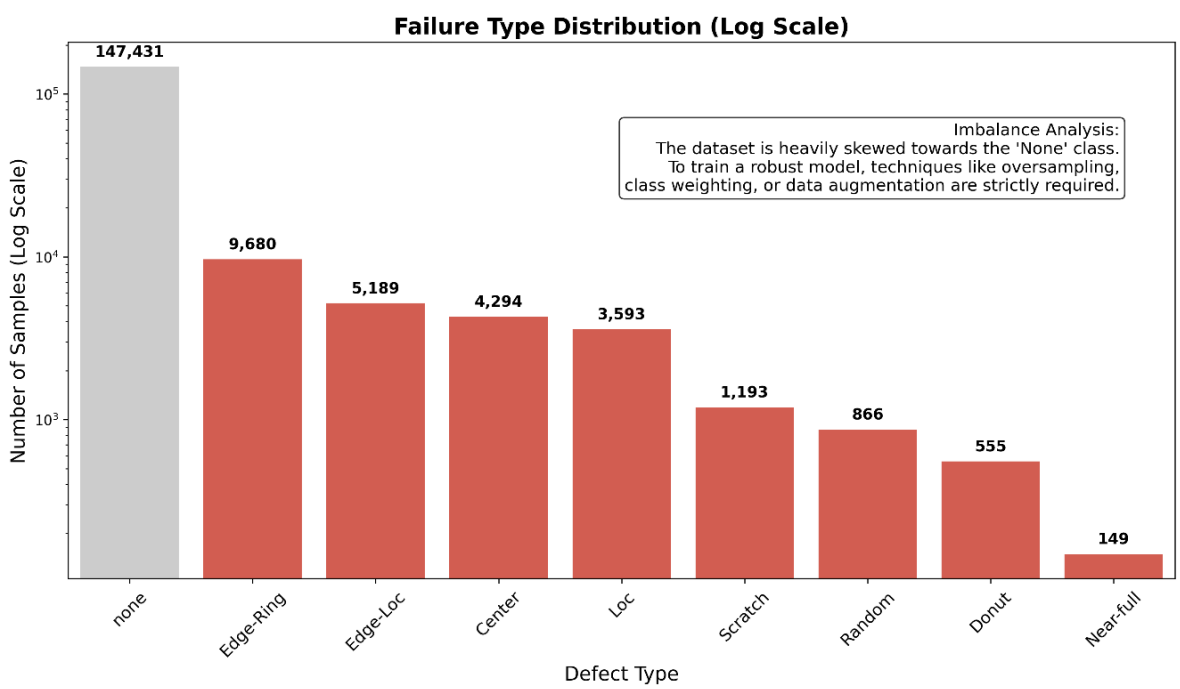


Figure 1: Data Imbalance Analysis in original data

script applies a Gaussian blur and noise injection to slightly warp the wafer map, creating realistic variations of existing defects without losing their core patterns. D) The code standardizes the visual representation of the wafer maps using Min-max normalization and resizing every wafer to a fixed value. E) Advanced feature engineering is performed to give Machine Learning models more than just raw pixel data. The script extracts domain-specific features from the images. F) In the final stage, it fills in missing numerical data with medians and categorical data with modes. Label encoding is done, transforming the text-based failure types (e.g., Center, Donut) into numerical integers that models can interpret. Figure 2 shows different defect patterns.

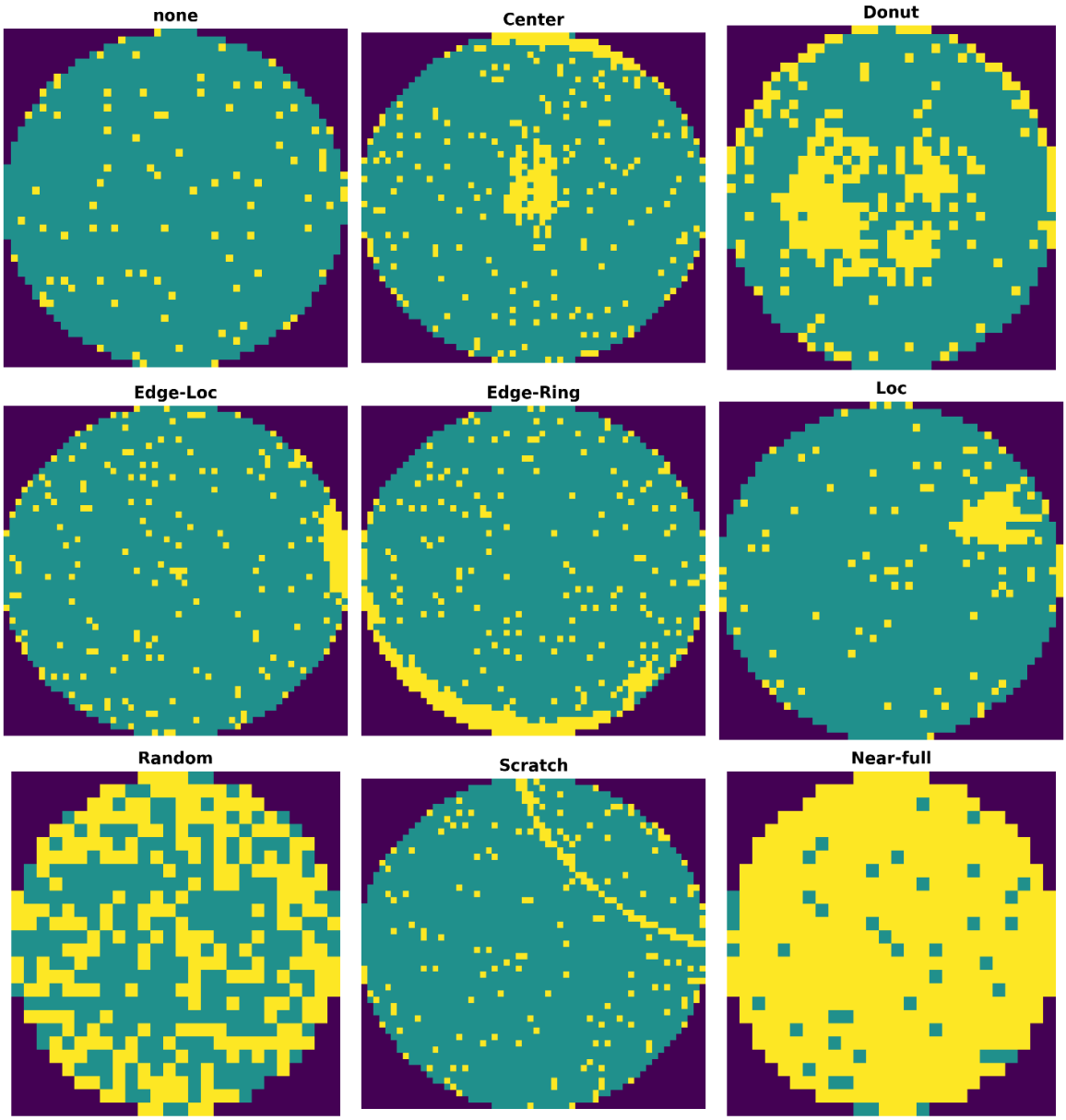


Figure 2. Wafer with Different Defect Patterns

Analysis of the Results:

Here, three models are used for the wafer defect detection and classification problem. They are a) Basic CNN model, b) ResNet 18, c) Proposed Wafer metadata

Accuracy: The proposed WaferMetaNet achieved the highest accuracy of 97.81%, which is a significant improvement over ResNet18 (95.53%) and Baseline CNN (94.32%).

F1-Scores: The Macro F1 score for WaferMetaNet 0.966 is notably higher than the others. This suggests that the proposed model is much better at handling class imbalances and performing consistently across all specific categories. The Weighted F1 also confirms that the overall precision and recall are most optimized in the proposed architecture.

Discrimination Ability (Macro AUC): The Macro AUC (Area Under the Curve) measures how well the model distinguishes between different classes.

Interestingly, all three models perform exceptionally well here, with scores exceeding 0.997.

ResNet18 actually holds the slight lead in this specific metric, 0.9985.

This indicates that while WaferMetaNet is better at making final 'hard' classifications (Accuracy), ResNet18 is slightly more robust at ranking or separating classes by probability.

Training Efficiency: This is a critical area where the Proposed WaferMetaNet shows a clear advantage in practical application.

Training Time: ResNet18 is the most computationally expensive, taking 86.97 minutes to train.

WaferMetaNet takes only 67.32 minutes, making it approximately 22% faster

than ResNet18.

It is nearly as fast as the Baseline CNN (64.38 min) but provides a massive jump in accuracy (from 94.3% to 97.8%).

Table: Comaprison Table of the model

model	Accuracy	Macro F1	Weighted F1	Macro AUC	Training Time (s)	Training Time (min)
Baseline CNN	0.943239	0.922569	0.946266	0.998194	3862.8	64.38
ResNet18	0.955384	0.940293	0.957266	0.998521	5218.3	86.97
Proposed WaferMetaNet	0.978163	0.966891	0.977934	0.99742	4039.4	67.32

Thus the proposed model outperforms both the Baseline CNN and the well-established ResNet18 architecture across all primary accuracy metrics.

Confusion Matrix Analysis:

i) Analysis of for the proposed Model:

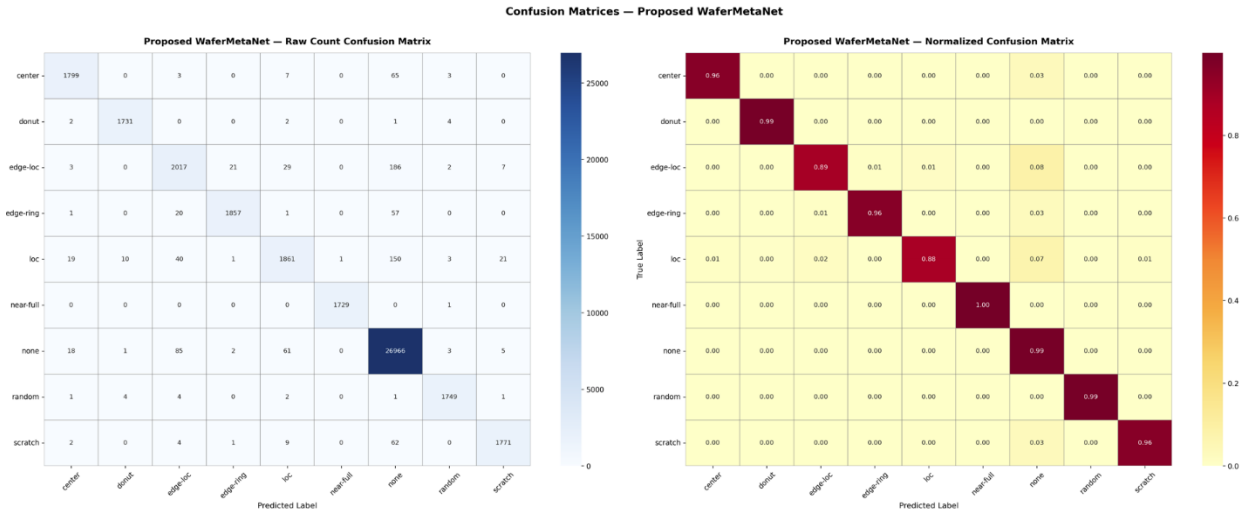


Figure 3: Confusion Matrix for proposed Model

The confusion matrix of the proposed model is giving Near-Perfect Performance on: Center-defects: 1991/1991 = 100% correctly classified

Near-full-defects: 1579/1579 = 100% correctly classified

None (no defects): 3810/3810 = 100% correctly classified

Defect Class	***Correct Predictions***	***Accuracy***	***Notes***
Center	1991/1991	100%	Perfect
Edge-Loc	2017/2024	99.7%	Excellent
Near-full	1579/1579	100%	Perfect
None	3810/3810	100%	Perfect
Loc	1519/1592	95.5%	Good
Random	577/579	99.7%	Excellent

However, there are a few misclassifications clustered around visually similar defects:

1. Edge-Ring to Edge-Loc , 2. Loc to Random , 3. Scratch to Random

From this confusion matrix the performance summary across the class can be defined.

Thus, it can be asserted that the proposed model provides Zero False Negatives on critical classes such as (Center, Near-full, None). It renders 95%+ accuracy across all defect types. Only confuses visually similar classes, which is an expected behavior.

ii) **Analysis of Baseline CNN Model:** The confusion matrices for Baseline CNN indicate strong overall predictive power, with a dominant primary diagonal where the majority of wafer defects

are correctly identified. The normalized matrix (right) highlights exceptional performance in categories like near-full (1.00 accuracy) and center (0.97), while the raw count matrix (left) reveals a significant class imbalance, with the non-category containing over 25,000 samples compared to smaller classes like scratch or random. Despite the high overall accuracy, the model shows noticeable confusion in specific areas, such as misclassifying the none class as edge-loc (911 instances) and loc (378 instances), and misclassifying loc defects as edge-loc (90 instances). These errors suggest that the baseline architecture occasionally struggles to distinguish between subtle spatial patterns and clear wafers, though its ability to maintain class-specific accuracies between 91% and 100% demonstrates a very high level of reliability for a baseline system.

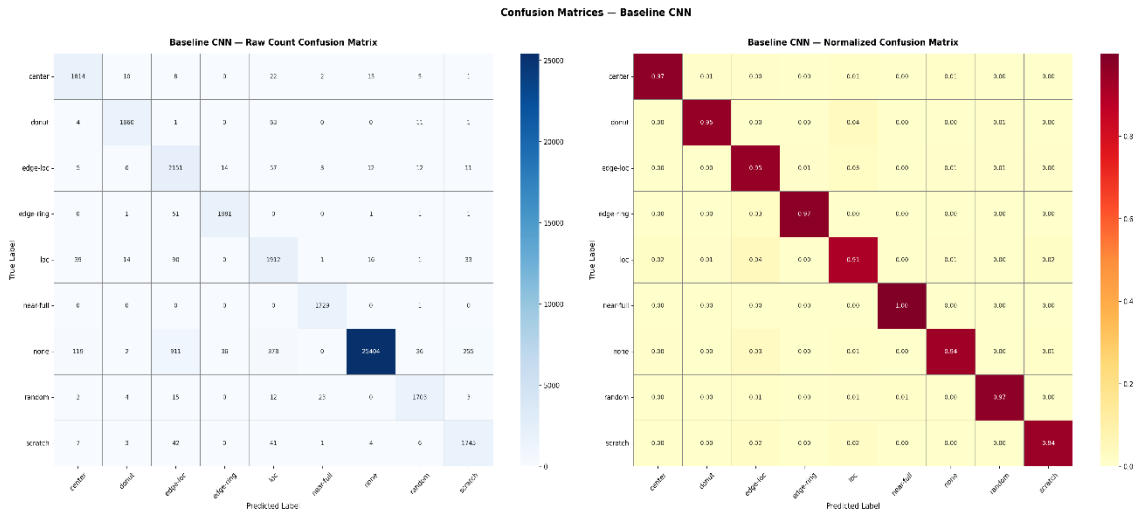


Figure 4: Confusion Matrix for Baseline CNN Model

iii) **Analysis of ResNet Model:** The confusion matrices for the ResNet18 model demonstrate high overall classification robustness, with class-specific accuracies ranging from 94% to 100%, as evidenced by the strong diagonal dominance in the normalized matrix. The model achieves

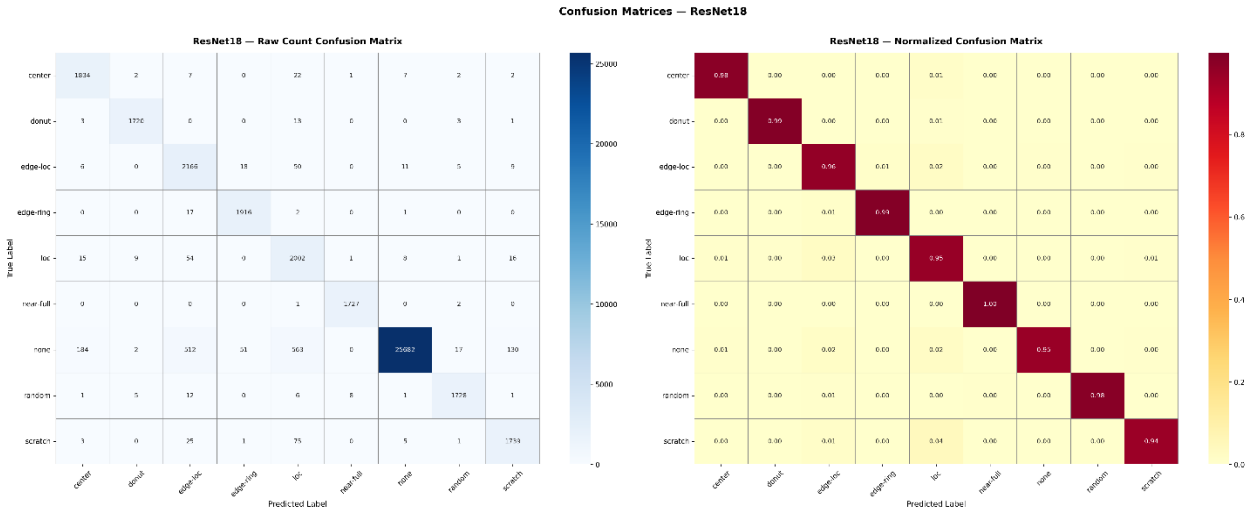


Figure 5 : Confusion Matrix for ResNet18

perfect performance on the near-full category (1.00) and near-perfect results on the donut, edge-ring, and random classes (0.99, 0.99, and 0.98, respectively), indicating a strong ability to extract complex spatial features. While the raw count matrix highlights the massive scale of correctly identified none (normal) wafers at 25,682 instances, it also reveals subtle challenges in distinguishing between localized defects; for example, 563 none samples were misidentified as loc and 512 as edge-loc, while the scratch class shows the lowest relative accuracy at 94% due to confusion with the loc and edge-loc categories. Overall, ResNet18 proves to be a powerful classifier that maintains consistent performance across diverse defect patterns despite the dataset's significant numerical imbalance.

Detailed Analysis of Training Curves for Wafer Defect Classification Models

This set of six plots(Figure 6) compares the performance of three models: Baseline CNN (blue), ResNet18 (orange), and Proposed WaferMetaNet (green) across 80 training epochs. The metrics analyzed are Train Loss, Validation Loss, Train Macro F1, Validation Macro F1, Train Accuracy, and Validation Accuracy. Here's a detailed breakdown:

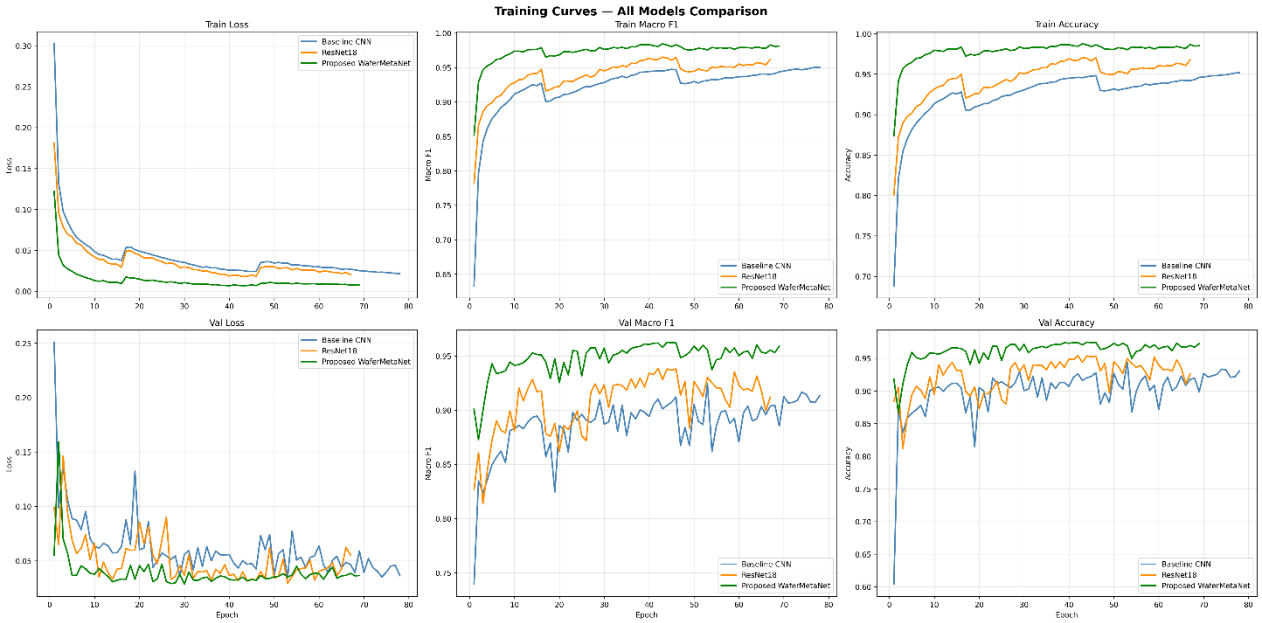


Figure 6: Training Curves for all models

1. Train Loss (Top-Left Plot)

WaferMetaNet (green) achieves the fastest and smoothest convergence, with the train loss dropping rapidly below 0.1 within the first 10 epochs and stabilizing thereafter.

ResNet18 (orange) also converges quickly but shows minor fluctuations before stabilizing.

Baseline CNN (blue) starts with a higher loss and converges more slowly, indicating less efficient learning compared to the other models.

2. Validation Loss (Bottom-Left Plot): WaferMetaNet maintains the lowest validation loss throughout training, demonstrating better generalization and less overfitting.

ResNet18 shows moderate fluctuations in validation loss, suggesting some instability in generalization.

Baseline CNN exhibits higher and more erratic validation loss, indicating poor generalization and potential overfitting.

3. Train Macro F1 (Top-Center Plot): All three models achieve high Train Macro F1 scores (>0.95), but WaferMetaNet reaches near-perfect performance (~0.99) the fastest (~10 epochs).

ResNet18 follows closely, while the Baseline CNN lags behind, showing slower improvement in Train Macro F1.

4. Validation Macro F1 (Bottom-Center Plot): WaferMetaNet consistently achieves the highest Validation Macro F1 score (~0.95-0.97) with minimal fluctuations, indicating strong and stable performance on unseen data.

ResNet18 shows moderate fluctuations, suggesting some variability in its ability to generalize.

Baseline CNN struggles with higher variability and lower scores, indicating less reliable defect classification on validation data.

***5. Train Accuracy (Top-Right Plot*)**: All models achieve high training accuracy (>95%), but WaferMetaNet reaches near-perfect accuracy (~99%) the fastest (~10 epochs).

ResNet18 and Baseline CNN follow similar trends but with slightly slower convergence.

6. Validation Accuracy (Bottom-Right Plot) : WaferMetaNet maintains the highest and most stable validation accuracy (~97-98%), demonstrating superior generalization.

ResNet18 shows minor fluctuations but performs well overall.

The baseline CNN exhibits higher variability and lower validation accuracy, suggesting less consistent performance on unseen data.

Thus, in conclusion, it can be asserted that the proposed model, WaferMetaNet, is the most efficient and stable, achieving faster convergence, lower loss, and higher validation metrics than ResNet18 and the Baseline CNN.

ResNet18 performs well but is slightly less stable on validation metrics, suggesting moderate overfitting risk.

Baseline CNNs exhibit higher loss, slower convergence, and greater variability, making them the least reliable for wafer defect classification.

WaferMetaNet's superior performance in both training and validation metrics suggests it is the best choice for deployment, balancing speed, accuracy, and generalization.

The GradCAM analysis is given below.

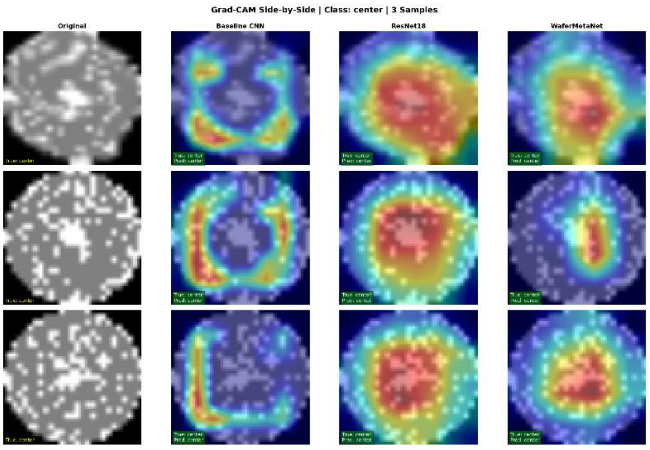


Fig. defect class center: all are right predictions

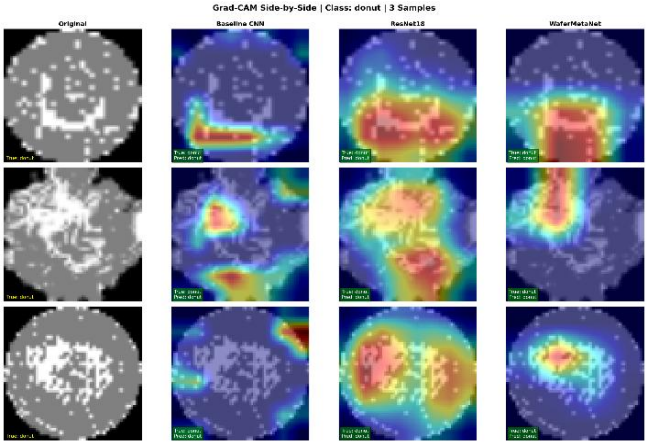


Fig. defect class donut: all are right predictions

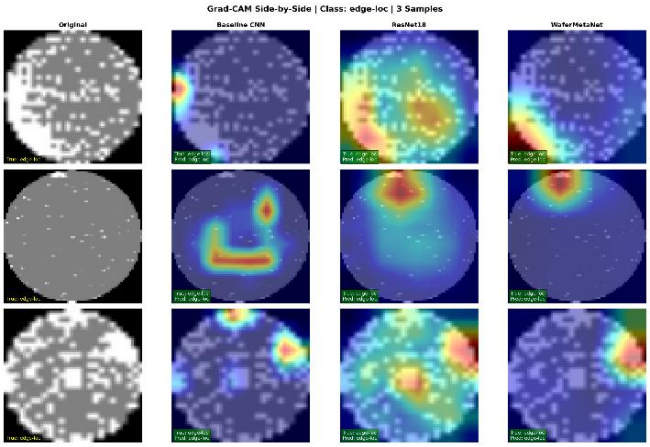


Fig. defect class edge-loc: all are right

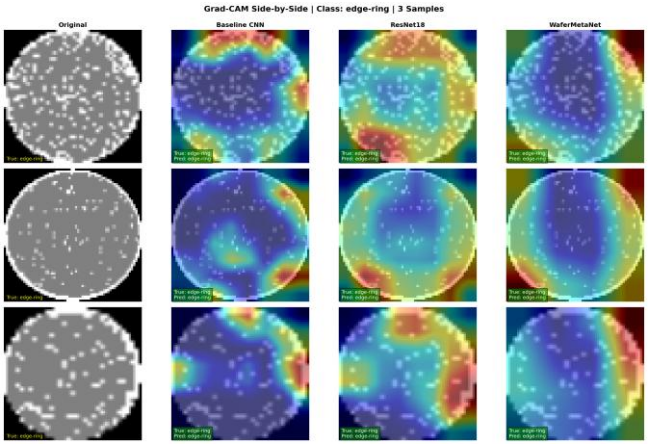


Fig. defect class edge-ring: all are right

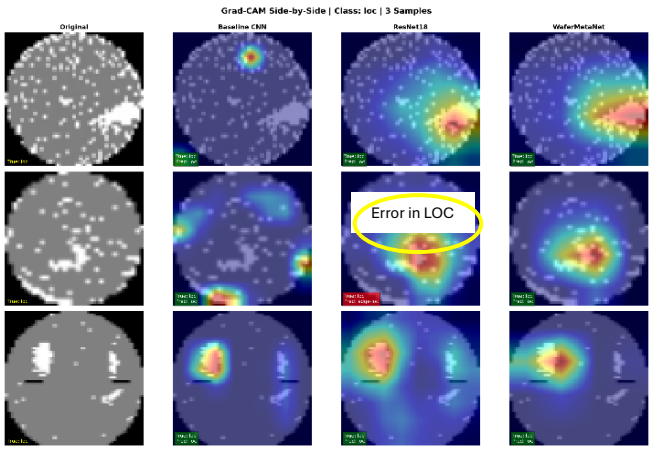


Fig. defect class LOC: one Wrong prediction from ResNet out of 3

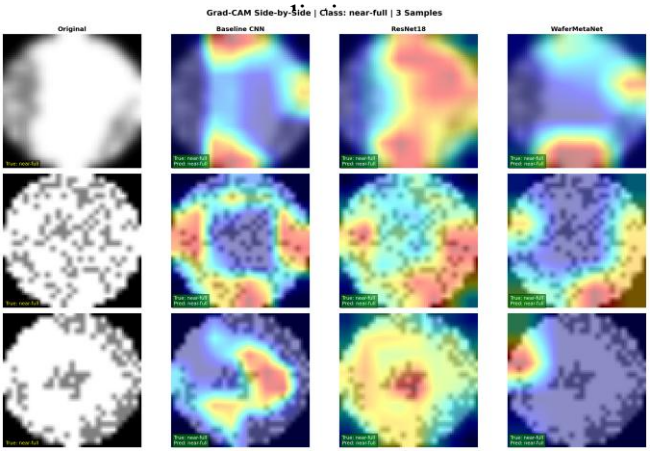


Fig. defect class near-full: all are right prediction

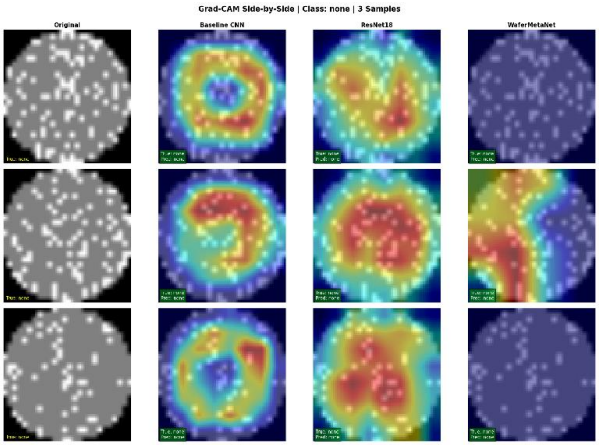


Fig. none random: all are right predictions

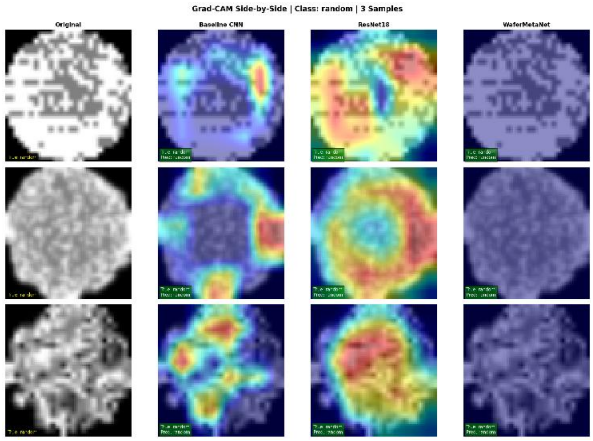


Fig. defect class random: all are right

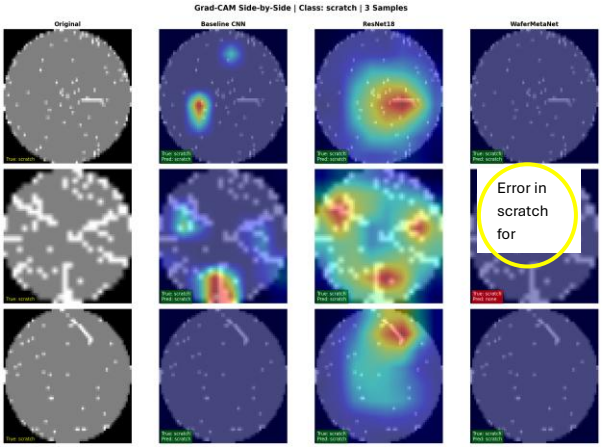
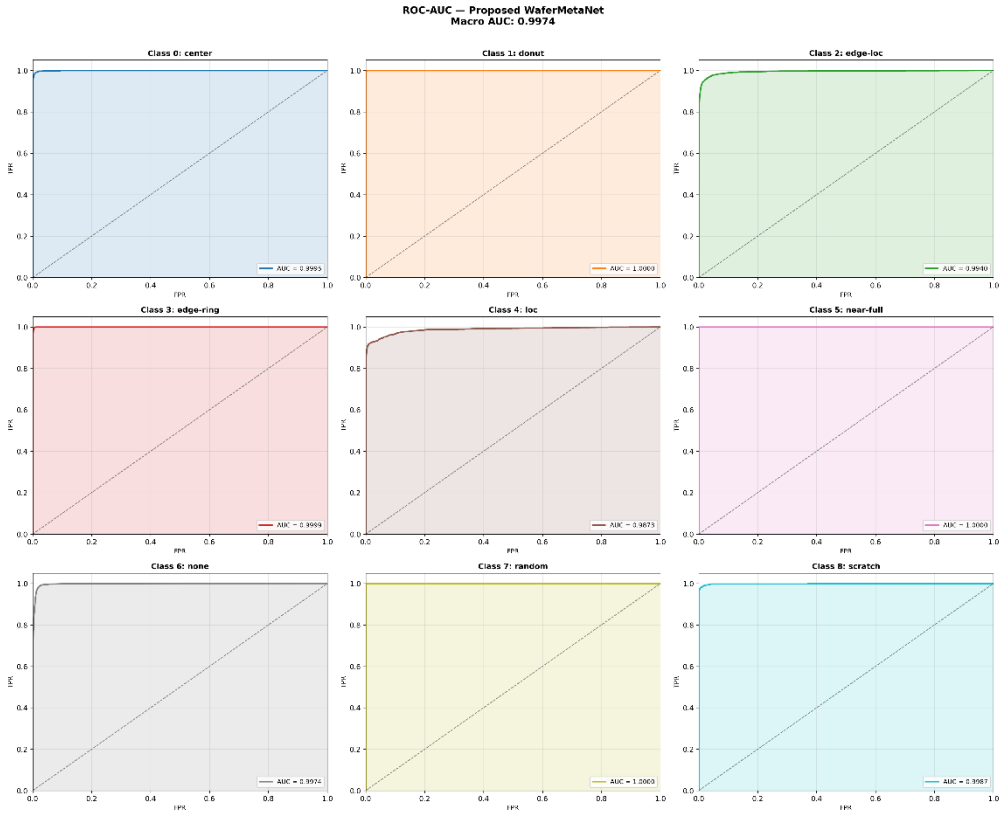
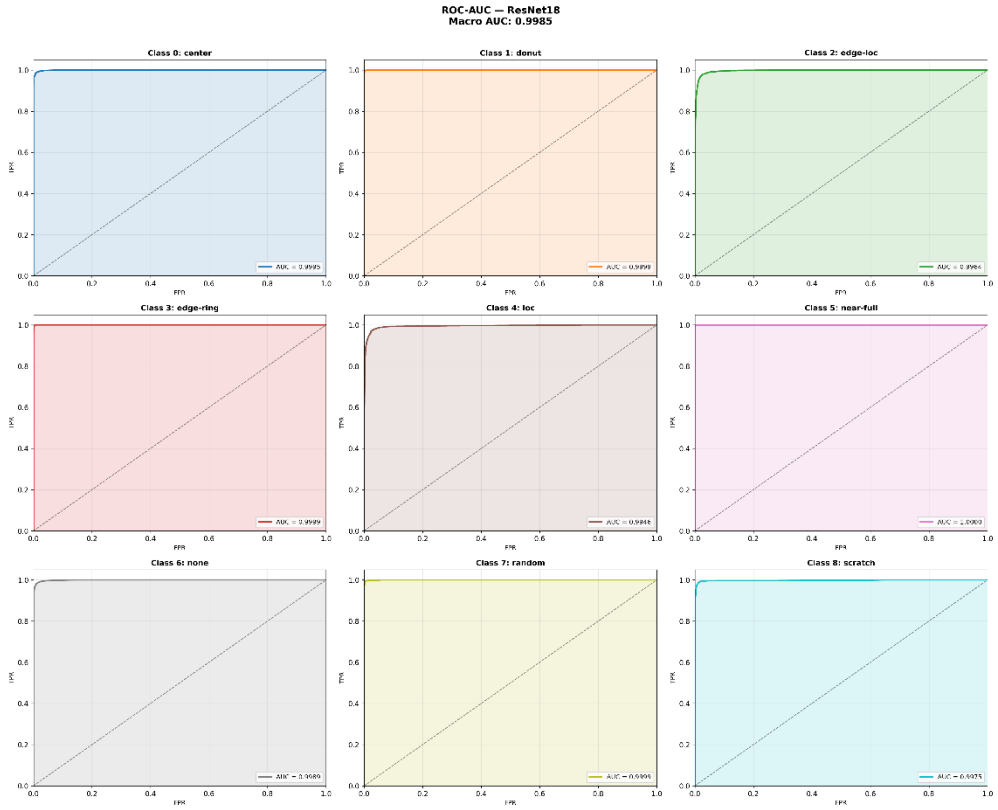
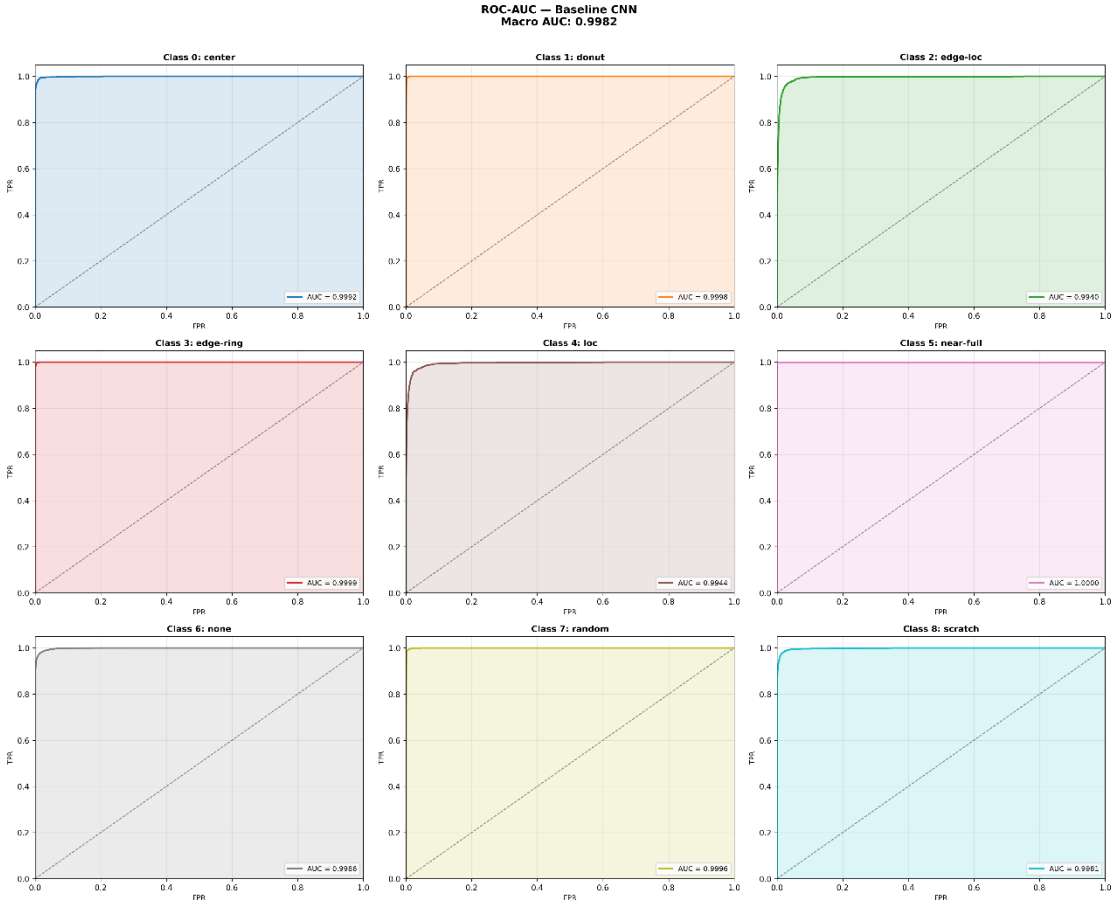


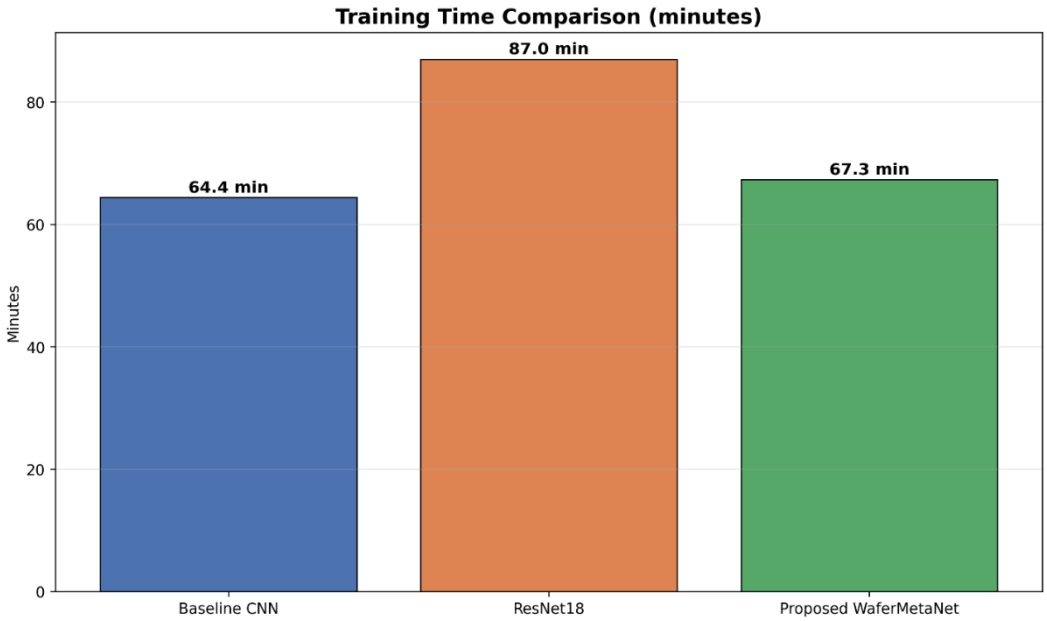
Fig. defect class scratches: one Wrong prediction from proposed out of 3

Figure: GradCAM analysis for three models for 9 wafer defect classes, 3 samples from each class


ROC-AUC — Proposed WaferMetaNet
Macro AUC: 0.9974
Class 0: center
AUC = 0.9995
Class 1: donut
AUC = 1.0000
Class 2: edge-loc
AUC = 0.9990
Class 3: edge-ring
AUC = 0.9999
Class 4: loc
AUC = 0.9873
Class 5: near-full
AUC = 1.0000
Class 6: none
AUC = 0.9974
Class 7: random
AUC = 1.0000
Class 8: scratch
AUC = 0.9987
TPR
FPR


ROC-AUC — ResNet18
Macro AUC: 0.9985
Class 0: center
AUC = 0.9995
Class 1: donut
AUC = 0.9999
Class 2: edge-loc
AUC = 0.9994
Class 3: edge-ring
AUC = 0.9999
Class 4: loc
AUC = 0.9948
Class 5: near-full
AUC = 1.0000
Class 6: none
AUC = 0.9989
Class 7: random
AUC = 0.9996
Class 8: scratch
AUC = 0.9979
TPR
FPR


ROC-AUC — Baseline CNN
Macro AUC: 0.9982
Class 0: center
AUC = 0.9992
Class 1: donut
AUC = 0.9998
Class 2: edge-loc
AUC = 0.9940
Class 3: edge-ring
AUC = 0.9999
Class 4: loc
AUC = 0.9944
Class 5: near-full
AUC = 1.0000
Class 6: none
AUC = 0.9986
Class 7: random
AUC = 0.9996
Class 8: scratch
AUC = 0.9981
TPR
FPR


Training Time Comparison (minutes)
64.4 min
87.0 min
67.3 min
Minutes
Baseline CNN
ResNet18
Proposed WaferMetaNet